

一、对下面给定的每一个周期离散时间信号，确定其傅里叶级数的系数。并画出每一组系数 a_k 的幅值和相位。

$$\text{a) } x(n) = \sin\left(\frac{n-1}{4}\pi\right)$$

$$x(n) = \sin\left(\frac{n-1}{4}\pi\right), \quad \frac{n-1}{4}\pi + 2\pi = \frac{1}{4}(n-1+8)\pi, \quad N = 8$$

$$\begin{aligned} x(n) &= \sin\left(\frac{n-1}{4}\pi\right) = \frac{1}{j2}(e^{j\frac{n-1}{4}\pi} - e^{-j\frac{n-1}{4}\pi}) = \frac{1}{j2}e^{-j\frac{1}{4}\pi}e^{j\frac{n}{4}\pi} - \frac{1}{j2}e^{j\frac{1}{4}\pi}e^{-j\frac{n}{4}\pi} \\ &= \frac{\sqrt{2}}{j4}(1-j)e^{j\frac{n}{4}\pi} - \frac{\sqrt{2}}{j4}(1+j)e^{-j\frac{n}{4}\pi} \end{aligned}$$

$$\begin{aligned} x(n) &= \sin\left(\frac{n-1}{4}\pi\right) = \frac{1}{j2}(e^{j\frac{n-1}{4}\pi} - e^{-j\frac{n-1}{4}\pi}) = \frac{1}{j2}e^{-j\frac{1}{4}\pi}e^{j\frac{n}{4}\pi} - \frac{1}{j2}e^{j\frac{1}{4}\pi}e^{-j\frac{n}{4}\pi} \\ &= \frac{\sqrt{2}}{j4}(1-j)e^{j\frac{1}{4}n\pi} - \frac{\sqrt{2}}{j4}(1+j)e^{j\frac{7}{4}n\pi} = \end{aligned}$$

$$a_1 = -\frac{\sqrt{2}}{4}(1+j) = \frac{1}{2}e^{j\frac{5}{4}\pi}, \quad a_7 = -\frac{\sqrt{2}}{4}(1-j) = \frac{1}{2}e^{j\frac{3}{4}\pi}$$

$$a_0 = a_2 = a_3 = a_4 = a_5 = a_6 = 0$$

$$\text{b) } x(n) = \cos\left(\frac{2n}{7}\pi\right) + \sin\left(\frac{2n}{3}\pi\right)$$

$$x(n) = \cos\left(\frac{2n}{7}\pi\right) + \sin\left(\frac{2n}{3}\pi\right), \quad \frac{2n}{7}\pi + 2\pi = 2\pi\left(\frac{n-7}{7}\right), \quad N_1 = 7$$

$$\frac{2n}{3}\pi + 2\pi = 2\pi\left(\frac{n-3}{3}\right), \quad N_2 = 3, \quad N = 21$$

$$\begin{aligned} x(n) &= \cos\left(\frac{2n}{7}\pi\right) + \sin\left(\frac{2n}{3}\pi\right) \\ &= \frac{1}{2}(e^{j\frac{2n}{7}\pi} + e^{-j\frac{2n}{7}\pi}) + \frac{1}{j2}(e^{j\frac{2n}{3}\pi} - e^{-j\frac{2n}{3}\pi}) \\ &= \frac{1}{2}e^{j\frac{6}{21}n\pi} + \frac{1}{2}e^{j\frac{15}{21}n\pi} + \frac{1}{j2}e^{j\frac{14}{21}n\pi} - \frac{1}{j2}e^{j\frac{7}{21}n\pi} \\ &= \frac{1}{2}e^{j\frac{6}{21}n\pi} - \frac{1}{j2}e^{j\frac{7}{21}n\pi} + \frac{1}{j2}e^{j\frac{14}{21}n\pi} + \frac{1}{2}e^{j\frac{15}{21}n\pi} \end{aligned}$$

$$a_6 = \frac{1}{2}, a_7 = \frac{j}{2}, a_{14} = -\frac{j}{2}, a_{15} = \frac{1}{2}$$

$$a_0 = \frac{1}{2}, a_7 = \frac{1}{2}e^{j\frac{1}{2}\pi}, a_{14} = \frac{1}{2}e^{j\frac{3}{2}\pi}, a_{15} = \frac{1}{2}, \quad a_0 \sim a_5, a_8 \sim a_{13}, a_{16} \sim a_{20} = 0$$

$$\text{c) } x(n) = \cos\left(\frac{11n}{4}\pi - \frac{\pi}{3}\right)$$

$$x(n) = \cos\left(\frac{11}{4}n\pi - \frac{1}{3}\pi\right) = \cos\left(\frac{3}{4}n\pi - \frac{1}{3}\pi + 2n\pi\right) = \cos\left(\frac{3}{4}n\pi - \frac{1}{3}\pi\right)$$

$$\frac{3n}{4}\pi - \frac{1}{3}\pi + 2r\pi = \frac{3}{4}\left(n + \frac{8}{3}r\right)\pi - \frac{1}{3}\pi, \quad N = \frac{8}{3}r = 8, \quad r = 3$$

$$x(n) = \cos\left(\frac{3}{4}n\pi - \frac{1}{3}\pi\right) = \frac{1}{2}\left(e^{j\left(\frac{3}{4}n\pi - \frac{1}{3}\pi\right)} + e^{-j\left(\frac{3}{4}n\pi - \frac{1}{3}\pi\right)}\right)$$

$$= \frac{1}{2}e^{-j\frac{1}{3}\pi}e^{j\frac{3}{4}n\pi} + \frac{1}{2}e^{j\frac{1}{3}\pi}e^{-j\frac{3}{4}n\pi} = \frac{1}{2}e^{-j\frac{1}{3}\pi}e^{j\frac{3}{4}n\pi} + \frac{1}{2}e^{j\frac{1}{3}\pi}e^{-j\frac{3}{4}n\pi}$$

$$a_3 = \frac{1}{2}e^{-j\frac{1}{3}\pi}, \quad a_5 = \frac{1}{2}e^{j\frac{1}{3}\pi}, \quad a_0 = a_1 = a_2 = a_4 = a_6 = a_7 = 0$$

d) $x(n)$ 以 6 为周期, 且 $x(n) = \left(\frac{1}{2}\right)^n, \quad -2 \leq n \leq 3$

$$x(n) = \left(\frac{1}{2}\right)^n, \quad -2 \leq n \leq 3, \quad N = 6$$

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=-N/2}^{N/2-1} x(n) e^{-j\frac{2\pi}{N}kn} = \frac{1}{6} \sum_{n=-2}^3 \left(\frac{1}{2}\right)^n e^{-j\frac{\pi}{3}kn} \\ &= \frac{1}{6} \left(4e^{j\frac{2\pi}{3}k} + 2e^{j\frac{\pi}{3}k} + 1 + \frac{1}{2}e^{-j\frac{\pi}{3}k} + \frac{1}{4}e^{-j\frac{2\pi}{3}k} + \frac{1}{8}e^{-j\pi k}\right) \\ &= \frac{2}{3}e^{j\frac{2\pi}{3}k} + \frac{1}{3}e^{j\frac{\pi}{3}k} + \frac{1}{6} + \frac{1}{12}e^{-j\frac{\pi}{3}k} + \frac{1}{24}e^{-j\frac{2\pi}{3}k} + \frac{1}{48}e^{-j\pi k} \end{aligned}$$

$$a_{-2} = \frac{2}{3}e^{-j\frac{4\pi}{3}} + \frac{1}{3}e^{-j\frac{2\pi}{3}} + \frac{1}{6} + \frac{1}{12}e^{j\frac{2\pi}{3}} + \frac{1}{24}e^{j\frac{4\pi}{3}} + \frac{1}{48}e^{j2\pi} = -\frac{3}{16}(2 - j\sqrt{3})$$

$$a_{-1} = \frac{2}{3}e^{-j\frac{2\pi}{3}} + \frac{1}{3}e^{-j\frac{\pi}{3}} + \frac{1}{6} + \frac{1}{12}e^{j\frac{\pi}{3}} + \frac{1}{24}e^{j\frac{2\pi}{3}} + \frac{1}{48}e^{j\pi} = -j\frac{7}{16}\sqrt{3}$$

$$a_0 = \frac{2}{3} + \frac{1}{3} + \frac{1}{6} + \frac{1}{12} + \frac{1}{24} + \frac{1}{48} = \frac{21}{16}$$

$$a_1 = \frac{2}{3}e^{j\frac{2\pi}{3}} + \frac{1}{3}e^{j\frac{\pi}{3}} + \frac{1}{6} + \frac{1}{12}e^{-j\frac{\pi}{3}} + \frac{1}{24}e^{-j\frac{2\pi}{3}} + \frac{1}{48}e^{-j\pi} = j\frac{7}{16}\sqrt{3}$$

$$a_2 = \frac{2}{3}e^{j\frac{4\pi}{3}} + \frac{1}{3}e^{j\frac{2\pi}{3}} + \frac{1}{6} + \frac{1}{12}e^{-j\frac{2\pi}{3}} + \frac{1}{24}e^{-j\frac{4\pi}{3}} + \frac{1}{48}e^{-j2\pi} = -\frac{3}{16}(2 + j\sqrt{3})$$

$$a_3 = \frac{2}{3}e^{j2\pi} + \frac{1}{3}e^{j\pi} + \frac{1}{6} + \frac{1}{12}e^{-j\pi} + \frac{1}{24}e^{-j2\pi} + \frac{1}{48}e^{-j3\pi} = \frac{7}{16}$$

$$\text{e) } x(n) = \sin\left(\frac{2n}{3}\pi\right)\cos\left(\frac{n}{2}\pi\right)$$

$$x(n) = \sin\left(\frac{2n}{3}\pi\right)\cos\left(\frac{n}{2}\pi\right), \quad N_1 = 3, N_2 = 4, \quad N = 12$$

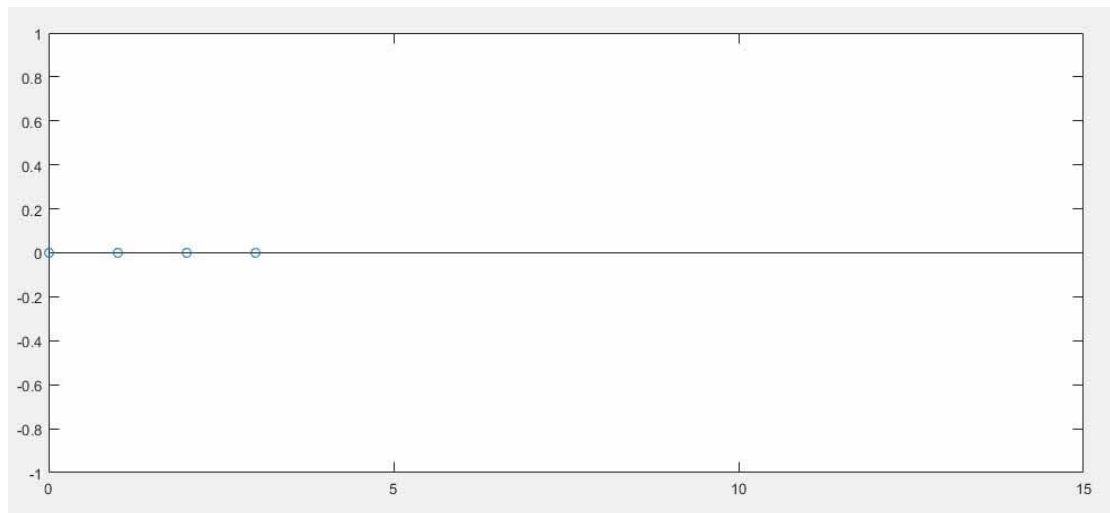
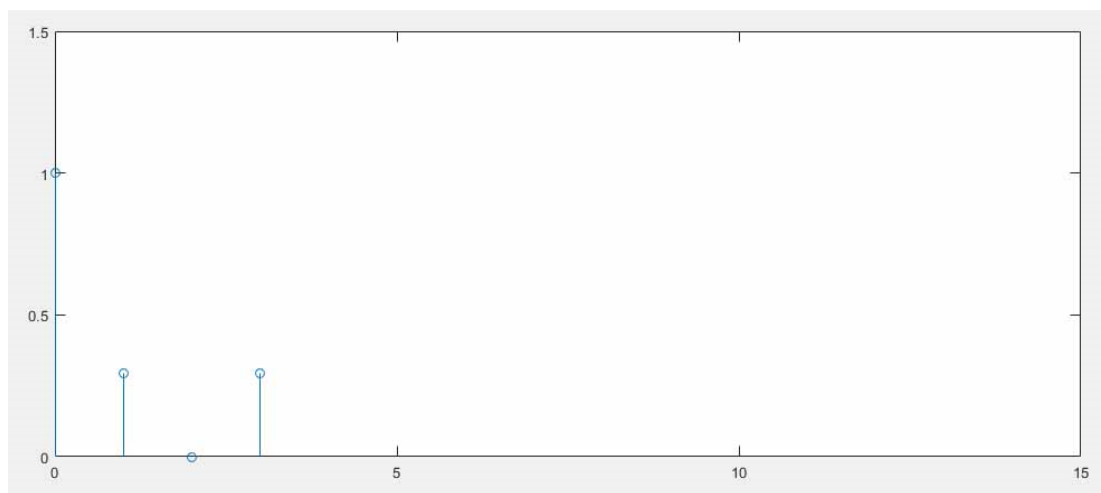
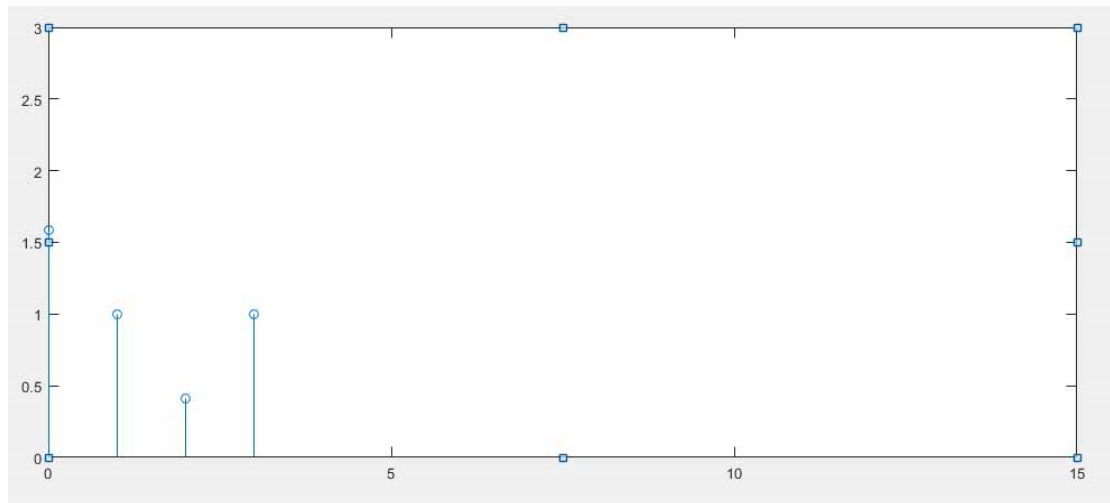
$$\begin{aligned} x(n) &= \sin\left(\frac{2n}{3}\pi\right)\cos\left(\frac{n}{2}\pi\right) = \frac{1}{j4}(e^{j\frac{2n}{3}\pi} - e^{-j\frac{2n}{3}\pi})(e^{j\frac{n}{2}\pi} + e^{-j\frac{n}{2}\pi}) \\ &= \frac{1}{j4}(e^{j(\frac{2}{3}+\frac{1}{2})n\pi} + e^{j(\frac{2}{3}-\frac{1}{2})n\pi} - e^{-j(\frac{2}{3}-\frac{1}{2})n\pi} - e^{-j(\frac{2}{3}+\frac{1}{2})n\pi}) \\ &= \frac{1}{j4}(e^{j\frac{7}{6}n\pi} + e^{j\frac{1}{6}n\pi} - e^{-j\frac{1}{6}n\pi} - e^{-j\frac{7}{6}n\pi}) = \frac{1}{j4}(e^{j\frac{7}{6}n\pi} + e^{j\frac{1}{6}n\pi} - e^{j\frac{11}{6}n\pi} - e^{j\frac{5}{6}n\pi}) \\ &= -\frac{j}{4}e^{j\frac{1}{6}n\pi} + \frac{j}{4}e^{j\frac{5}{6}n\pi} - \frac{j}{4}e^{j\frac{7}{6}n\pi} + \frac{j}{4}e^{j\frac{11}{6}n\pi} \\ &= \frac{1}{4}e^{-j\frac{1}{2}\pi}e^{j\frac{1}{6}n\pi} + \frac{1}{4}e^{j\frac{1}{2}\pi}e^{j\frac{5}{6}n\pi} + \frac{1}{4}e^{-j\frac{1}{2}\pi}e^{j\frac{7}{6}n\pi} + \frac{1}{4}e^{j\frac{1}{2}\pi}e^{j\frac{11}{6}n\pi} \\ a_1 &= \frac{1}{4}e^{-j\frac{1}{2}\pi}, \quad a_5 = \frac{1}{4}e^{j\frac{1}{2}\pi}, \quad a_7 = \frac{1}{4}e^{-j\frac{1}{2}\pi}, \quad a_{11} = \frac{1}{4}e^{j\frac{1}{2}\pi}, \\ a_0 &= a_2 = a_3 = a_4 = a_6 = a_8 = a_9 = a_{10} = 0 \end{aligned}$$

$$\text{f) } x(n) \text{ 以 } 4 \text{ 为周期, 且 } x(n) = 1 - \sin\left(\frac{n}{4}\pi\right), \quad 0 \leq n \leq 3$$

$$x(n) = 1 - \sin\left(\frac{n}{4}\pi\right), \quad 0 \leq n \leq 3$$

$$x(0) = 1, \quad x(1) = 1 - \frac{\sqrt{2}}{2}, \quad x(2) = 0, \quad x(3) = 1 - \frac{\sqrt{2}}{2}$$

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi}{N}kn} = \frac{1}{4} \sum_{n=0}^3 x(n)e^{-j\frac{\pi}{2}kn} \\ &= \frac{1}{4} \left(1 + \frac{2-\sqrt{2}}{2}e^{-j\frac{\pi}{2}k} + \frac{2-\sqrt{2}}{2}e^{-j\frac{3\pi}{2}k}\right) \\ a_0 &= \frac{3-\sqrt{2}}{4}, \quad a_1 = \frac{1}{4}, \quad a_2 = \frac{-1+\sqrt{2}}{4}, \quad a_3 = \frac{1}{4} \end{aligned}$$



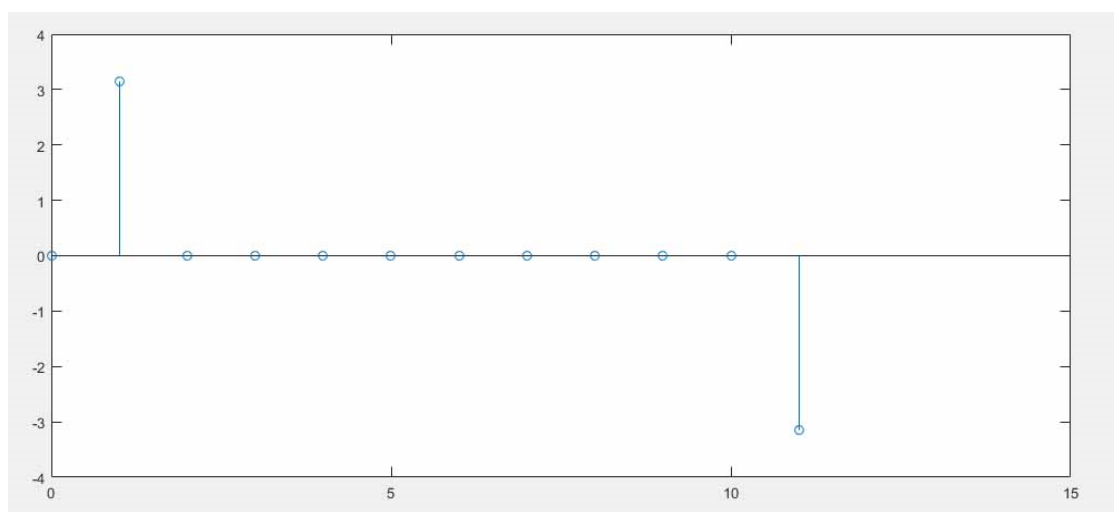
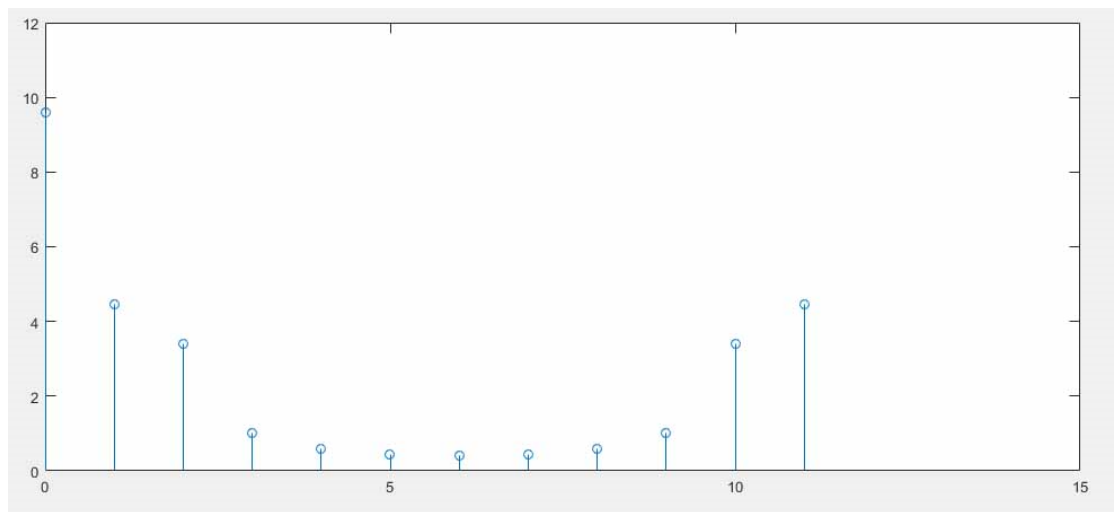
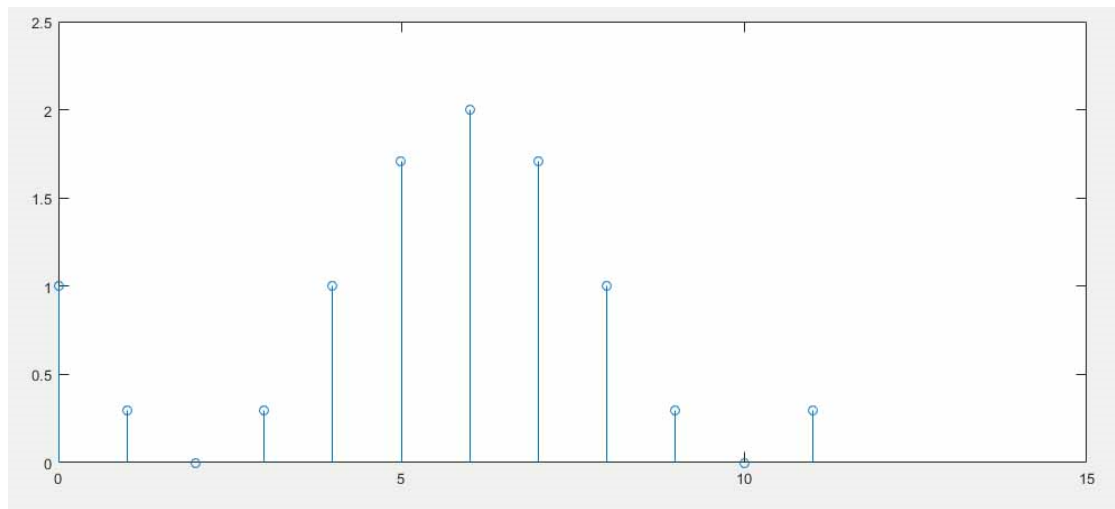
g) $x(n)$ 以 12 为周期, 且 $x(n)=1-\sin(\frac{n}{4}\pi)$, $0 \leq n \leq 11$

$$x(n)=1-\sin(\frac{n}{4}\pi), \quad 0 \leq n \leq 11$$

$$\begin{aligned} a_k &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} kn} = \frac{1}{12} \sum_{n=0}^{11} x(n) e^{-j \frac{\pi}{6} kn} \\ &= \frac{1}{12} \sum_{n=0}^{11} (1 - \sin(\frac{n}{4}\pi)) e^{-j \frac{\pi}{6} kn} = \frac{1}{12} \sum_{n=0}^{11} (1 - \frac{1}{j2} (e^{j \frac{n}{4}\pi} - e^{-j \frac{n}{4}\pi})) e^{-j \frac{\pi}{6} kn} \\ &= \frac{1}{12} (\sum_{n=0}^{11} e^{-j \frac{\pi}{6} kn} - \frac{1}{j2} \sum_{n=0}^{11} e^{j \frac{n}{4}\pi} e^{-j \frac{\pi}{6} kn} + \frac{1}{j2} \sum_{n=0}^{11} e^{-j \frac{n}{4}\pi} e^{-j \frac{\pi}{6} kn}) \\ &= \frac{1}{12} \sum_{n=0}^{11} e^{-j \frac{\pi}{6} kn} - \frac{1}{j24} \sum_{n=0}^{11} e^{j \frac{3-2k}{12} n\pi} + \frac{1}{j24} \sum_{n=0}^{11} e^{-j \frac{3+2k}{12} n\pi}, \end{aligned}$$

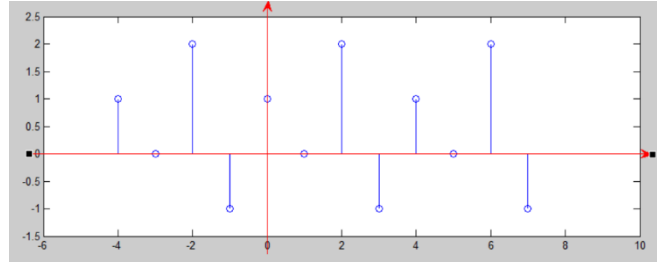
$$S_n = a_0 \frac{1-q^{n+1}}{1-q}$$

$$\begin{aligned} &= \frac{1}{12} \frac{1-e^{-j \frac{\pi}{6} 12k}}{1-e^{-j \frac{\pi}{6} k}} - \frac{1}{j24} \frac{1-e^{j \frac{3-2k}{12} 12\pi}}{1-e^{j \frac{3-2k}{12} \pi}} + \frac{1}{j24} \frac{1-e^{-j \frac{3+2k}{12} 12\pi}}{1-e^{-j \frac{3+2k}{12} \pi}} \\ &= \frac{1}{12} \frac{1-e^{-j 2k\pi}}{1-e^{-j \frac{\pi}{6} k}} - \frac{1}{j24} \frac{1-e^{j(3-2k)\pi}}{1-e^{j \frac{3-2k}{12} \pi}} + \frac{1}{j24} \frac{1-e^{-j(3+2k)\pi}}{1-e^{-j \frac{3+2k}{12} \pi}} \\ a_k|_{k \neq 0} &= -\frac{1}{j24} \frac{1-(-1)}{1-e^{j \frac{3-2k}{12} \pi}} + \frac{1}{j24} \frac{1-(-1)}{1-e^{-j \frac{3+2k}{12} \pi}} = \frac{1}{j12} (\frac{1}{1-e^{j \frac{3-2k}{12} \pi}} + \frac{1}{1-e^{-j \frac{3+2k}{12} \pi}}) \\ &= \frac{1}{j12} \frac{2-e^{j \frac{3-2k}{12} \pi} - e^{-j \frac{3+2k}{12} \pi}}{1-e^{j \frac{3-2k}{12} \pi} - e^{-j \frac{3+2k}{12} \pi} + e^{-j \frac{k}{3} \pi}} = \frac{1}{j12} \frac{2-\sqrt{2}e^{-j \frac{k}{6} \pi}}{1-\sqrt{2}e^{-j \frac{k}{6} \pi} + e^{-j \frac{k}{3} \pi}} \\ a_0 &= 1 - \frac{j}{12} \quad a_k = -\frac{j}{12} \frac{2-\sqrt{2}e^{-j \frac{k}{6} \pi}}{1-\sqrt{2}e^{-j \frac{k}{6} \pi} + e^{-j \frac{k}{3} \pi}}, \quad k=1, 2, \dots, 11 \end{aligned}$$



二、研究下图中所画的信号 $x(n]$ ，该信号是周期的，且周期 $N=4$ 。信号可以用离散时间傅里叶级数表示为：

$$x(n) = \sum_{k=0}^3 a_k e^{jk(\frac{2\pi}{4})n}$$



a) 写出这 4 个方程，利用求解含有 4 个未知数的 4 个方程的任何一种标准方法，直接解这个方程。（首先，务必把上面的复指数函数化简为最简单的形式）

b) 通过利用离散时间傅里叶级数分析公式：

$$a_k = \frac{1}{4} \sum_{n=0}^3 x(n) e^{-jk(\frac{2\pi}{4})n}$$

直接计算 a_k ，来验证上述答案。

$$x(0)=1, \quad x(1)=0, \quad x(2)=2, \quad x(3)=-1$$

$$x(n) = \sum_{k=0}^3 a_k e^{jk(\frac{2\pi}{4})n} = a_0 + a_1 e^{j\frac{\pi}{2}n} + a_2 e^{j\pi n} + a_3 e^{j\frac{3\pi}{2}n}$$

$$\begin{cases} x(0) = a_0 + a_1 + a_2 + a_3 \\ x(1) = a_0 + a_1 e^{j\frac{\pi}{2}} + a_2 e^{j\pi} + a_3 e^{j\frac{3\pi}{2}} \\ x(2) = a_0 + a_1 e^{j\pi} + a_2 e^{j2\pi} + a_3 e^{j3\pi} \\ x(3) = a_0 + a_1 e^{j\frac{3\pi}{2}} + a_2 e^{j3\pi} + a_3 e^{j\frac{9\pi}{2}} \end{cases} \Rightarrow \begin{cases} a_0 + a_1 + a_2 + a_3 = 1 \\ a_0 + ja_1 - a_2 - ja_3 = 2 \\ a_0 - a_1 + a_2 - a_3 = 0 \\ a_0 - ja_1 - a_2 + ja_3 = -1 \end{cases}$$

$$a_0 = 1, \quad a_1 = -\frac{1}{4}(1+j), \quad a_2 = \frac{1}{2}, \quad a_3 = -\frac{1}{4}(1-j)$$

$$x(0)=1, \quad x(1)=0, \quad x(2)=2, \quad x(3)=-1$$

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn} = \frac{1}{4} (1 + 2e^{-j\pi k} - e^{-j\frac{3\pi}{2}k})$$

$$= \frac{1}{4} (1 + 2(-1)^k + (-j)^k)$$

$$a_0 = 1, \quad a_1 = -\frac{1}{4}(1+j), \quad a_2 = \frac{1}{2}, \quad a_3 = -\frac{1}{4}(1-j)$$

三、 在下面每一小题中，我们规定了周期为 8 的一个周期性信号的傅里叶级数的系数。试对每一种情况，确定信号 $x(n)$

$$1) \quad a_k = \cos\left(\frac{k}{4}\pi\right) + \sin\left(\frac{3k}{4}\pi\right)$$

$$\begin{aligned} x(n) &= \sum_{k=0}^7 a_k e^{j\frac{2\pi}{8}kn} = \sum_{k=0}^7 \left(\cos\frac{\pi}{4}k + \sin\frac{3\pi}{4}k\right) e^{j\frac{\pi}{4}kn} \\ &= \frac{1}{2} \sum_{k=0}^7 (e^{j\frac{\pi}{4}k} + e^{-j\frac{\pi}{4}k}) e^{j\frac{\pi}{4}kn} + \frac{1}{j2} \sum_{k=0}^7 (e^{j\frac{3\pi}{4}k} - e^{-j\frac{3\pi}{4}k}) e^{j\frac{\pi}{4}kn} \\ &= \frac{1}{2} \sum_{k=0}^7 (e^{j\frac{\pi}{4}(n+1)k} + e^{j\frac{\pi}{4}(n-1)k}) + \frac{1}{j2} \sum_{k=0}^7 (e^{j\frac{\pi}{4}(n+3)k} - e^{j\frac{\pi}{4}(n-3)k}) \\ &= \frac{1}{2} \left(\frac{1 - e^{j\frac{\pi}{4}(n+1)8}}{1 - e^{j\frac{\pi}{4}(n+1)}} + \frac{1 - e^{j\frac{\pi}{4}(n-1)8}}{1 - e^{j\frac{\pi}{4}(n-1)}} \right) + \frac{1}{j2} \left(\frac{1 - e^{j\frac{\pi}{4}(n+3)8}}{1 - e^{j\frac{\pi}{4}(n+3)}} + \frac{1 - e^{j\frac{\pi}{4}(n-3)8}}{1 - e^{j\frac{\pi}{4}(n-3)}} \right) \\ &= \frac{1}{2} \left(\frac{1 - e^{j2\pi(n+1)}}{1 - e^{j\frac{\pi}{4}(n+1)}} + \frac{1 - e^{j2\pi(n-1)}}{1 - e^{j\frac{\pi}{4}(n-1)}} \right) + \frac{1}{j2} \left(\frac{1 - e^{j2\pi(n+3)}}{1 - e^{j\frac{\pi}{4}(n+3)}} + \frac{1 - e^{j2\pi(n-3)}}{1 - e^{j\frac{\pi}{4}(n-3)}} \right) \end{aligned}$$

$$x(n) = 0, \quad n = 0, 2, 4, 6$$

$$x(1) = \frac{1}{2} \left(\frac{1 - e^{j2\pi(n-1)}}{1 - e^{j\frac{\pi}{4}(n-1)}} \right) \bigg|_{n=1} = 4$$

$$x(3) = \frac{1}{j2} \left(\frac{1 - e^{j2\pi(n-3)}}{1 - e^{j\frac{\pi}{4}(n-3)}} \right) \bigg|_{n=3} = -j4 = 4e^{-j\frac{\pi}{2}}$$

$$x(5) = \frac{1}{j2} \left(\frac{1 - e^{j2\pi(n+3)}}{1 - e^{j\frac{\pi}{4}(n+3)}} \right) \bigg|_{n=5} = -j4 = 4e^{-j\frac{\pi}{2}}$$

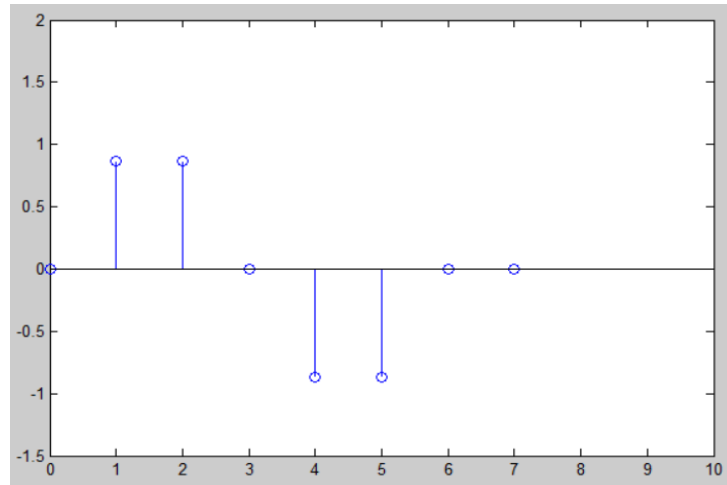
$$x(7) = \frac{1}{2} \left(\frac{1 - e^{j2\pi(n+1)}}{1 - e^{j\frac{\pi}{4}(n+1)}} \right) \bigg|_{n=7} = 4$$

$$a_k, \quad 0 \leq k \leq 7$$

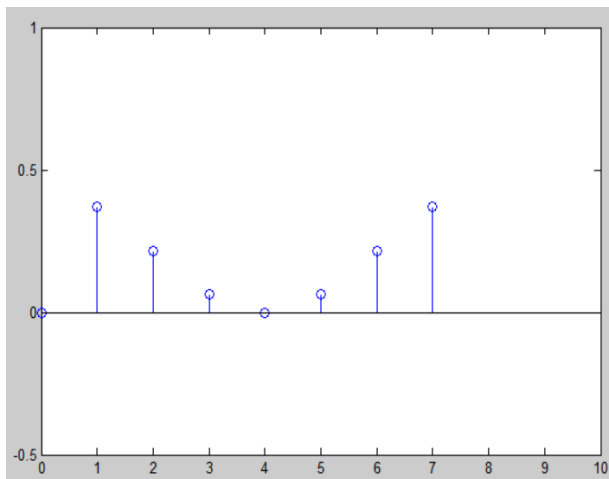
$$|x(n)|, \quad 0 \leq n \leq 7$$

$$\varphi(x(n)), \quad 0 \leq n \leq 7$$

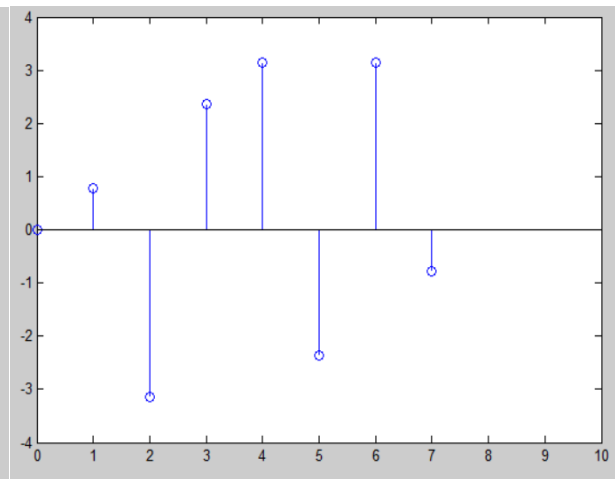
$$2) \ a_k = \begin{cases} \sin(\frac{k}{3}\pi) & 0 \leq k \leq 6 \\ 0 & k = 7 \end{cases} \quad x(n) = \sum_{k=0}^7 a_k e^{j\frac{2\pi}{8}kn} = \sum_{k=0}^6 \sin(\frac{k}{3}\pi) e^{j\frac{2\pi}{8}kn}$$



$a_k, \ 0 \leq k \leq 7$



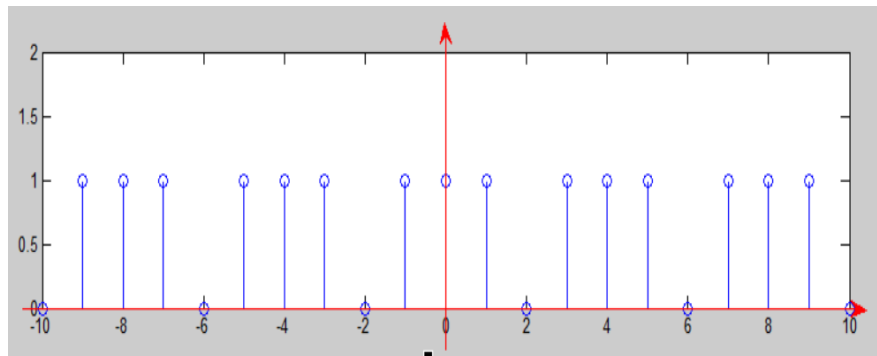
$|x(n)|, \ 0 \leq n \leq 7$



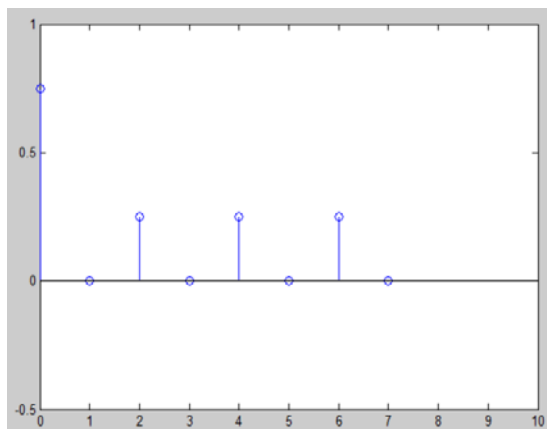
$\phi(x(n)), \ 0 \leq n \leq 7$

3) a_k 如下图所示。

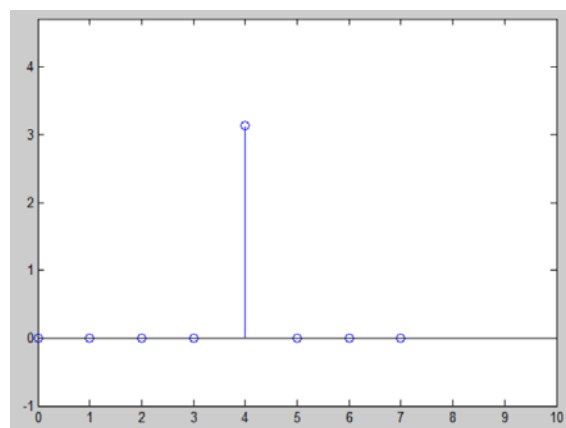
$$x(n) = \sum_{k=0}^7 a_k e^{j\frac{2\pi}{8}kn}$$



$$a_k, \quad 0 \leq k \leq 7$$

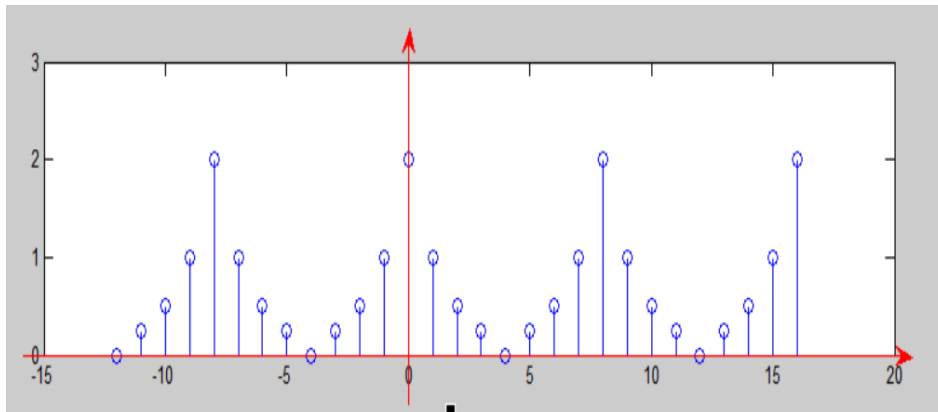


$$|x(n)|, \quad 0 \leq n \leq 7$$

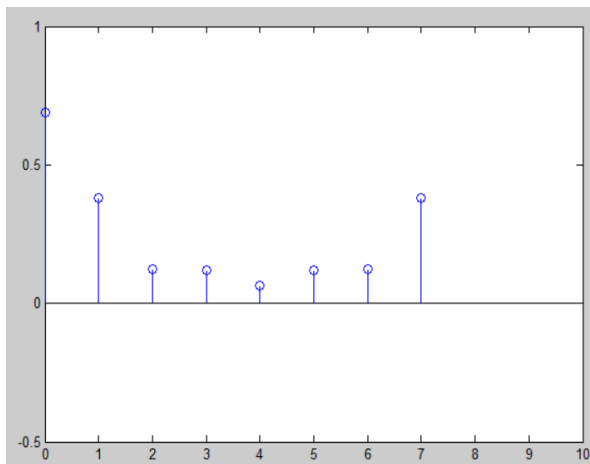


$$\phi(x(n)), \quad 0 \leq n \leq 7$$

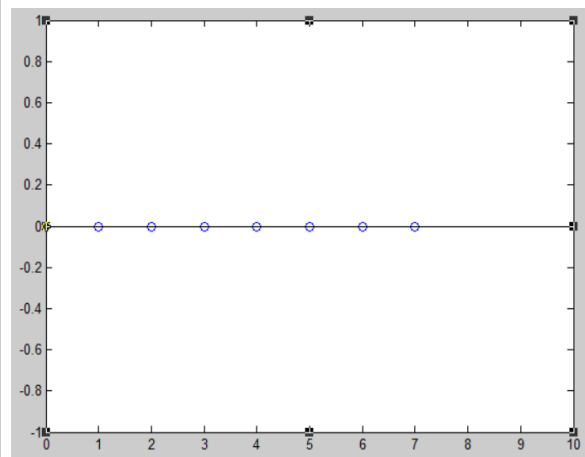
4) a_k 如下图所示。
$$x(n) = \sum_{k=0}^7 a_k e^{j\frac{2\pi}{8}kn}$$



$a_k, \quad 0 \leq k \leq 7$



$|x(n)|, \quad 0 \leq n \leq 7$



$\phi(x(n)), \quad 0 \leq n \leq 7$

四、 设 $x(n)$ 是个周期序列，其周期为 N ，傅里叶级数表示为：

$$x(n) = \sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{N})n} \quad (1)$$

a) 下列每个信号的傅里叶技术的系数都能用上式中的 a_k 来表示，请导出这些表达式。

1) $x(n-n_0)$

$$\begin{aligned} x(n) &= \sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{N})n}, \quad a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n) e^{-jk(\frac{2\pi}{N})n} \\ a'_k &= \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n-n_0) e^{-jk(\frac{2\pi}{N})n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n-n_0) e^{-jk(\frac{2\pi}{N})(n-n_0)} e^{-jk(\frac{2\pi}{N})n_0} \\ &= e^{-jk(\frac{2\pi}{N})n_0} \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n-n_0) e^{-jk(\frac{2\pi}{N})(n-n_0)} = a_k e^{-jk(\frac{2\pi}{N})n_0} \end{aligned}$$

2) $x(n) - x(n-1)$

$$\begin{aligned} x(n) &= \sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{N})n}, \quad a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n) e^{-jk(\frac{2\pi}{N})n} \\ a'_k &= \frac{1}{N} \sum_{n=-\infty}^{\infty} \{x(n) - x(n-1)\} e^{-jk(\frac{2\pi}{N})n} = \{a_k - \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n-1) e^{-jk(\frac{2\pi}{N})(n-1)} e^{-jk\frac{2\pi}{N}}\} \\ &= \{a_k - e^{-jk\frac{2\pi}{N}} \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n-1) e^{-jk(\frac{2\pi}{N})(n-1)}\} = a_k (1 - e^{-jk\frac{2\pi}{N}}) \end{aligned}$$

3) $x^*(-n)$

$$\begin{aligned} x(n) &= \sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{N})n}, \quad a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n) e^{-jk(\frac{2\pi}{N})n} \\ a'_k &= \frac{1}{N} \sum_{n=-\infty}^{\infty} \{x^*(-n)\} e^{-jk(\frac{2\pi}{N})n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} \{x(-n)\} e^{-jk(\frac{2\pi}{N})(-n)} \}^* \\ &= a_k^* \end{aligned}$$

4) $x(n) - x(n - \frac{N}{2})$ N 是偶数

$$\begin{aligned}
x(n) &= \sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{N})n}, \quad a_k = \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n) e^{-jk(\frac{2\pi}{N})n} \\
a'_k &= \frac{1}{N} \sum_{n=-\infty}^{\infty} \{x(n) - x(n - \frac{N}{2})\} e^{-jk(\frac{2\pi}{N})n} = \{a_k - \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n - \frac{N}{2}) e^{-jk(\frac{2\pi}{N})(n - \frac{N}{2})} e^{-jk\frac{2\pi}{N}\frac{N}{2}}\} \\
&= \{a_k - e^{-jk\pi} \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n - \frac{N}{2}) e^{-jk(\frac{2\pi}{N})(n - \frac{N}{2})}\} = a_k (1 - (-1)^k) \\
&= \begin{cases} 2 & k = 2p + 1 \\ 0 & k = 2p \end{cases}, \quad p = 0, 1, \dots, \frac{N}{2} - 1
\end{aligned}$$

b) 假若 N 为偶数，且上述 (1) 式中的 $x(n)$ 满足：

$$x(n) = -x(n + \frac{N}{2}) \quad \text{对所有的 } n$$

证明：对所有的偶整数 k ， $a_k = 0$

$$\begin{aligned}
x(n) &= \sum_{k=-\infty}^{\infty} a_k e^{jk(\frac{2\pi}{N})n} \\
a_k &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-jk(\frac{2\pi}{N})n} = \frac{1}{N} \left(\sum_{n=0}^{\frac{N}{2}-1} x(n) e^{-jk(\frac{2\pi}{N})n} + \sum_{n=\frac{N}{2}}^{N-1} x(n) e^{-jk(\frac{2\pi}{N})n} \right) \\
&= \frac{1}{N} \left(\sum_{n=0}^{\frac{N}{2}-1} x(n) e^{-jk(\frac{2\pi}{N})n} + \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2}) e^{-jk(\frac{2\pi}{N})(n + \frac{N}{2})} \right) \\
&= \frac{1}{N} \left(\sum_{n=0}^{\frac{N}{2}-1} x(n) e^{-jk(\frac{2\pi}{N})n} - e^{-jk\pi} \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{-jk(\frac{2\pi}{N})n} \right) \\
&= \frac{1}{N} \left(\sum_{n=0}^{\frac{N}{2}-1} x(n) e^{-jk(\frac{2\pi}{N})n} - (-1)^k \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{-jk(\frac{2\pi}{N})n} \right) \\
&= \begin{cases} 0 & k = 2p \\ \frac{1}{N} \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{-jk(\frac{2\pi}{N})n} & k = 2p + 1 \end{cases}
\end{aligned}$$

五、 一个 LTI 系统的单位冲激响应为：

$$h(n) = \left(\frac{1}{2}\right)^n$$

$$h(n) = \left(\frac{1}{2}\right)^n$$

$$H(\omega) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n} = \frac{1}{1 - \frac{1}{2}e^{-j\omega}} = \frac{2}{2 - e^{-j\omega}}$$

$$|H(\omega)| = \frac{1}{\sqrt{2 - \cos \omega}}, \quad \varphi(H(\omega)) = \tan^{-1} \frac{\sin \omega}{\cos \omega - 2}$$

对下列输入，求该系统输出 $y(n)$ 的傅里叶级数表达式。

a) $x(n) = \sin\left(\frac{3n}{4}\pi\right)$

$$x(n) = \sin\left(\frac{3}{4}n\pi\right), \quad N = 8, \quad \omega = \frac{3}{4}\pi$$

$$y(n) = \frac{1}{\sqrt{2 - \cos \omega}} \sin\left(\frac{3}{4}n\pi + \varphi\right), \quad \varphi = \tan^{-1} \frac{\sin \frac{3}{4}\pi}{\cos \frac{3}{4}\pi - 2} = \tan^{-1} \frac{2\sqrt{2} - 1}{9} \approx 0.2$$

b) $x(n) = \sum_{k=-\infty}^{+\infty} \delta(n - 4k)$

$$x(n) = \sum_{m=-\infty}^{+\infty} \delta(n - 4m), \quad N = 4$$

$$X(\omega) = 2\pi \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{k}{4}\pi\right)$$

$$\begin{aligned} Y(\omega) &= X(\omega)H(\omega) = 2\pi \sum_{k=-\infty}^{+\infty} \delta\left(\omega - \frac{k}{4}\pi\right) \frac{2}{2 - e^{-j\omega k}} \\ &= 2\pi \sum_{k=-\infty}^{+\infty} \frac{2}{2 - e^{-j\frac{k}{4}\omega\pi}} = 2\pi \sum_{k=-\infty}^{+\infty} \frac{1}{1 - \frac{1}{2}e^{-j\frac{k}{4}\omega\pi}} \end{aligned}$$

c) $x(n)$ 是一个周期信号，其周期为 7，且： $x(n) = \begin{cases} 1 & n = 0, \pm 1 \\ 0 & n = \pm 2, \pm 3 \end{cases}$

$$x(n) = \begin{cases} 1 & n = 0, \pm 1 \\ 0 & n = \pm 2, \pm 3 \end{cases}, \quad N = 6$$

$$\begin{aligned} a_k &= \frac{1}{6} \sum_{n=-3}^{+2} x(n) e^{-j\frac{\pi}{3}kn} = \frac{1}{6} \sum_{n=-1}^{+1} e^{-j\frac{\pi}{3}kn} = \frac{1}{6} (e^{j\frac{\pi}{3}k} + 1 + e^{-j\frac{\pi}{3}k}) \\ &= \frac{1}{6} \left(1 + 2\cos\left(\frac{k}{3}\pi\right)\right) \end{aligned}$$

$$a_{-3} = -\frac{1}{6}, \quad a_{-2} = 0, \quad a_{-1} = \frac{1}{3}, \quad a_0 = \frac{1}{2}, \quad a_1 = \frac{1}{3}, \quad a_2 = 0$$

$$\begin{aligned}
 x(n) &= \sum_{k=-3}^2 a_k e^{j\frac{\pi}{3}kn} = -\frac{1}{6}e^{-j\pi n} + \frac{1}{3}e^{-j\frac{\pi}{3}n} + \frac{1}{2} + \frac{1}{3}e^{j\frac{\pi}{3}n} \\
 &= -\frac{1}{6}(-1)^n + \frac{2}{3}\cos\left(\frac{\pi}{3}n\right) + \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 y(n) &= \sum_{k=-3}^2 a_k A_k e^{j\frac{\pi}{3}kn} e^{j\varphi_k} \\
 &= -\frac{1}{6}A_{-3}e^{j\varphi_{-3}}e^{-j\frac{\pi}{3}3n} + \frac{1}{3}A_{-1}e^{j\varphi_{-1}}e^{-j\frac{\pi}{3}n} + \frac{1}{2}A_0e^{j\varphi_0} + \frac{1}{3}A_1e^{j\varphi_1}e^{j\frac{\pi}{3}n}
 \end{aligned}$$

$$A_{-3} = \frac{1}{\sqrt{2 - \cos \pi}} = \frac{\sqrt{3}}{3}, \quad \varphi_{-3} = \tan^{-1} \frac{\sin \pi}{\cos \pi - 2} = 0$$

$$A_{-1} = \frac{1}{\sqrt{2 - \cos(-\frac{\pi}{3})}} = \frac{\sqrt{6}}{3}, \quad \varphi_{-1} = \tan^{-1} \frac{\sin(-\frac{\pi}{3})}{\cos(-\frac{\pi}{3}) - 2} = \tan^{-1} \frac{\sqrt{3}}{3} = 0.523$$

$$A_0 = \frac{1}{\sqrt{2 - \cos 0}} = 1, \quad \varphi_0 = \tan^{-1} \frac{\sin 0}{\cos 0 - 2} = 0$$

$$A_1 = \frac{1}{\sqrt{2 - \cos(\frac{\pi}{3})}} = \frac{\sqrt{6}}{3}, \quad \varphi_1 = \tan^{-1} \frac{\sin(\frac{\pi}{3})}{\cos(\frac{\pi}{3}) - 2} = -\tan^{-1} \frac{\sqrt{3}}{3} = -0.523$$

$$\begin{aligned}
 y(n) &= -\frac{\sqrt{3}}{18}(-1)^n + \frac{\sqrt{6}}{9}e^{j0.522}e^{-j\frac{\pi}{3}n} + \frac{1}{2} + \frac{\sqrt{6}}{9}e^{-j0.522}e^{j\frac{\pi}{3}n} \\
 &= \frac{1}{2} + \frac{2\sqrt{6}}{9}\cos\left(\frac{\pi}{3}n - 0.522\right) - \frac{\sqrt{3}}{18}(-1)^n
 \end{aligned}$$